LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Hostel Maintenance & Issue Ticketing System

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Domain	Campus & Academic Operations
Actors	Hostel Resident, Maintenance Warden

1. Scenario

The Hostel Maintenance & Issue Ticketing System allows hostel residents to report maintenance problems such as electrical, plumbing, furniture, room, and other hostel facility issues. Residents can submit and track tickets, while the Maintenance Warden can view reported issues, assign maintenance work, update ticket status, and mark issues as resolved.

2. Written Justification

Architectural Choice: We chose **Layered Architecture** for the Hostel Maintenance & Issue Ticketing System.

Two Reasons:

1. **Clear separation of responsibilities:** The Resident Portal and Warden interface can be kept in the Presentation Layer, ticket creation, assignment and status processing can be handled in the Business Logic Layer, and resident/ticket records can be handled in the Data Layer. This means a change to the user interface does not require changes to database operations.
2. **Suitable for the hostel maintenance workflow:** The system follows a straightforward centralized workflow—residents submit maintenance issues, the system stores the tickets, the Maintenance Warden views and assigns them, and the ticket status is updated until resolution. Layered Architecture provides a simple structure for these operations without the operational complexity of independently deployed services.

Security Advantage: The Authentication Service can be separated from the user interface and used by the business layer for authentication and role-based authorization. This allows the system to restrict residents to appropriate ticket operations while giving the Maintenance Warden permission to assign tickets and update maintenance status.

Performance Benefit: The layered structure keeps ticket-processing logic and database access organized in dedicated layers. For common operations such as submitting a ticket, retrieving ticket status, or updating a ticket, the system can follow a direct Presentation → Business Logic → Data flow, reducing unnecessary communication and processing between unrelated parts of the system.

3. Components

Component	Purpose
Resident Portal	Report maintenance issues and view/track tickets.
Authentication Service	Handle login and role-based authorization.
Ticket Management	Create, assign, update, and resolve maintenance tickets.
Maintenance Management	Allow the Maintenance Warden to view issues, assign work, and update status.
Ticket Database	Store resident details, ticket records, status, assignment, and resolution data.

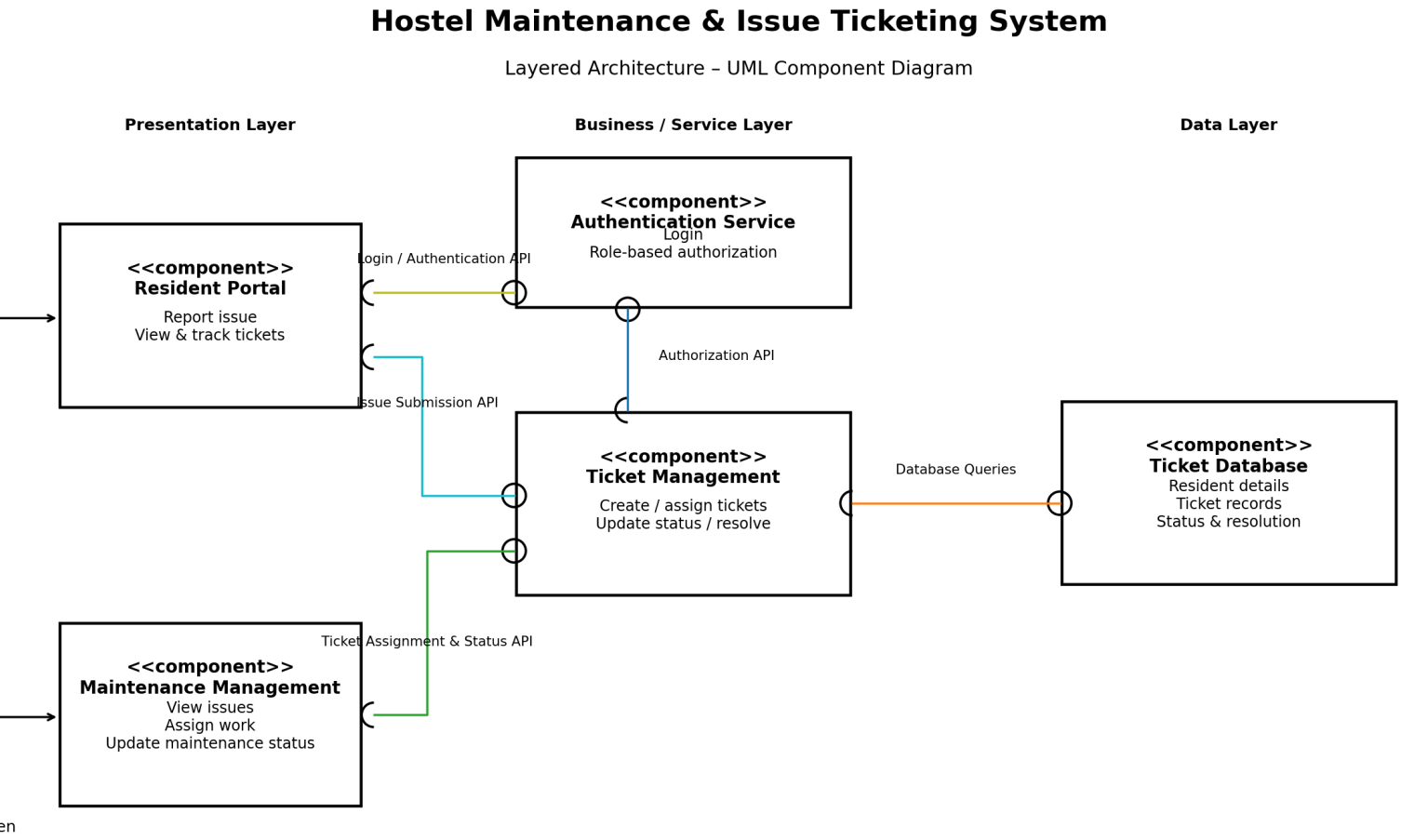
4. Interfaces

Components	Interface
Resident Portal ↔ Authentication Service	Login / Authentication API
Resident Portal ↔ Ticket Management	Issue Submission API
Ticket Management ↔ Authentication Service	Authorization API

Components	Interface
Ticket Management ↔ Ticket Database	Database Queries
Maintenance Management ↔ Ticket Management	Ticket Assignment & Status API

5. UML Component Diagram

Layered Architecture – 5 Components, 5 Interfaces



6. UML Notation Used

- Component notation is used for all five system components.
- Provided interfaces use ball/circle notation.
- Required interfaces use socket/semicircle notation.
- Assembly connectors connect required interfaces to provided interfaces.
- Hostel Resident and Maintenance Warden are shown as the two actors.

7. Submission Checklist

- ■ Architectural Choice stated.
- ■ Two scenario-related reasons provided.
- ■ Security Advantage explained.
- ■ Performance Benefit explained.
- ■ At least 5 components included.
- ■ At least 4 interfaces required; 5 interfaces included.
- ■ UML component diagram included.